

Executive Summary & Architectural Purpose

SmritiNER is a standardized, multilingual digital therapeutic designed for elderly patients with Mild Cognitive Impairment (MCI) and early-stage dementia. This document provides the definitive mathematical, clinical, and software specifications for: (1) the micro-telemetry and Bayesian Item Response Theory (2PL IRT) merit formulas of all 10 active games, (2) the multi-game daily aggregation formula that strictly excludes unplayed games, (3) the on-device edge ML classification pipeline trained on Washington University's OASIS-2 longitudinal cohort, and (4) the comprehensive clinical PDF report generation engine.

1. Mathematical Foundations & Game Merit Formulas

Each game in SmritiNER evaluates a distinct neuropsychological function grounded in validated clinical assessments (e.g. Cambridge CANTAB, MoCA, WAIS-IV, WMS-IV). Intra-game telemetry records raw response accuracy, millisecond-precision reaction latency, motor jitter with a 400ms exponential tremor debounce filter, and assistance degradation.

General Bayesian 2PL Item Response Theory (IRT) Foundation

For any item / trial i with difficulty b_i and discrimination a_i , the patient's probability of success given latent ability $\theta \in [-3.0, +3.0]$ is:

$$P(Y_i = 1 \mid \theta) = c_i + (1 - c_i) / [1 + \exp(-a_i(\theta - b_i))]$$

where c_i is pseudo-guessing (0 for open recall, $1/k$ for k -alternative forced choice). Latent ability θ updates sequentially via Bayesian expected a posteriori (EAP) updates.

# / Game ID	Clinical Standard	DSM-5 Domain	Merit & Ability Formulas
1. Memory Match memory-match	Cambridge CANTAB PAL (Paired Associates)	Visual Associative Memory	• Accuracy: $\text{Acc} = (\text{Pairs} / (\text{Flips} / 2)) \times 100\%$ • Bayesian θ: $\theta = \text{clamp}(-3.0, +3.0, (\text{Acc} - 50)/25 - (\text{Errors} \times 0.2/\text{Pairs}))$ • Merit Score: $0.70 \times \text{Acc} + 0.30 \times [(\theta + 3)/6 \times 100] - \text{Perseverations}$
2. Sequence Recall sequence-recall	Corsi Block-Tapping Forward/Reverse Span	Attention & Working Memory	• Effective Span: $\text{Span}_{\text{eff}} = \text{MaxForward} + 0.5 \times \text{ReverseSpan}$ • Deliberation: $\text{RT} = \text{avg}[\text{max}(\theta, \Delta t - 400\text{ms tremor filter})]$ • Bayesian θ: $\theta = (\text{Span}_{\text{eff}} - 3.5) \times 0.65 + ((\text{Acc} - 70)/30) \times 0.40$ • Est. MoCA WM: $\min(5, \max(1, \text{round}(0.8 \times \text{Span}_{\text{eff}} + 0.5 \times \theta)))$
3. Smriti Haat word-recall	MoCA Delayed Verbal Recall (5 Words)	Delayed Verbal Recall	• Retention Rate: $(\text{Correct Words} / 5) \times 100\%$ • Sensitivity Index: $d' = Z(\text{HitRate}) - Z(\text{FalseAlarmRate})$ • Bayesian θ: $\theta = 0.5 \times d' + (\text{Retention} - 60)/20$ • Merit Score: $\text{clamp}(50, 99, \text{round}(\text{Retention} \times 0.85 + \text{Bonus}_d))$
4. Jigsaw Puzzle jigsaw-puzzle	WAIS-IV Block Design & Visuomotor Praxis	Visuoconstructional Praxis	• Placement Acc: $(\text{Solved} / (\text{Pieces} + \text{Misplacements})) \times 100\%$ • Processing Speed: $\text{SpeedRatio} = \text{ExpectedTime} / \max(\text{ActualTime}, 10)$ • Bayesian θ: $(\text{Acc} - 65)/20 + 0.35 \times \ln(\text{SpeedRatio}) - 0.15 \times \text{Assists}$ • Praxis Merit: $\text{round}(0.60 \times \text{Acc} + 0.40 \times \theta_{\text{norm}})$
5. Number Recall number-recall	WAIS Digit Span & MMSE Serial 7s	Working Memory & Concentration	• Scaled Score: $\text{CorrectTrials} \times 2.5 - (\text{BackspaceCorrections} \times 0.5)$ • Hesitation Ratio: $(\sum \text{KeystrokePause} > 3000\text{ms}) / \text{TotalDuration}$ • Bayesian θ: $\theta = (\text{WAIS}_{\text{Scaled}} - 10) / 3.5$ • Merit Score: $\text{round}(\text{WAIS}_{\text{Scaled}} \times 5.2 + \theta_{\text{norm}} \times 0.4)$
6. What Changed? what-changed	Visual Change Blindness Binding Paradigm	Visual Attention & Change Binding	• Sensitivity A': $0.5 + [(\text{Hit} - \text{FA})(1 + \text{Hit} - \text{FA})] / [4 \times \text{Hit}(1 - \text{FA})]$ • Bayesian θ: $\theta = (A' - 0.75) \times 4.0$ • Merit Score: $\text{round}(0.75 \times \text{Accuracy} + 0.25 \times \theta_{\text{norm}})$

7. Brain Story brain-story	WMS-IV Narrative Logical Memory & Reminiscence	Narrative Memory & Language	• Narrative Recall Acc: (Correct Probes / Total Questions) × 100% • Autonomy Score: 100 - (AudioReplays × 10) - (VisualHints × 15) • Bayesian θ: (Acc - 70)/15 + (Autonomy ≥ 80 ? +0.25 : -0.25) • Merit Score: round(0.65 × Acc + 0.25 × θ_{norm} + 0.10 × Autonomy)
8. Where Am I? where-am-i	Retrosplenial Orientation & Landmark Wayfinding	Topographical Orientation	• Topographical Acc: (Correct Landmarks / Total Trials) × 100% • Hesitation Penalty: $\Delta_{Hesitation} =$ (MeanDeliberation > 7000ms ? 5 : 0) • Bayesian θ: (Acc - 70)/20 - ($\Delta_{Hesitation}$ / 10) • Merit Score: clamp(50, 98, round(Acc × 0.85 + θ_{norm} × 0.15))
9. Odd One Out odd-one-out	Wisconsin Card Sorting Test (WCST) Abstraction	Executive Function & Flexibility	• Abstraction Acc: (Correct Selections / Total Trials) × 100% • Rule Perseveration Rate: FailedCategoryShiftTaps / TotalErrorTaps • Bayesian θ: ((Acc - 65)/18) - (1.2 × PerseverationRate) • Merit Score: round(0.70 × Acc + 0.30 × θ_{norm} - 4 × Perseverations)
10. Pattern Recall pattern-recall	Matrix Working Memory & Reasoning Span	Visuospatial Working Memory	• Matrix Binding Acc: (Correct Cells / Target Cells) × 100% • Bayesian θ: ((Acc - 60)/18) + (LevelReached × 0.25) • Deliberation: Tremor filtered tap latency (Δt - 400ms) • Merit Score: round(0.65 × Acc + 0.25 × θ_{norm} + 0.10 × LevelSpan)

2. Daily Multi-Game Aggregation & Unplayed Game Exclusion System

The composite cognitive score displayed on the caregiver dashboard and exported into clinical reports combines telemetry **strictly from the games that were actually recommended and played on that specific day**. A critical flaw in standard consumer apps is assigning a 0 score or default placeholder to unplayed prescribed exercises, creating false alarms of cognitive decline. SmritiNER resolves this with a mathematically honest formulation.

CLINICAL INVARIANT — THE UNPLAYED GAME EXCLUSION RULE:

- If a patient is prescribed 4 exercises and completes only 2 today (e.g., *Smriti Haat* and *Jigsaw Puzzle*), the daily score is calculated **strictly across those 2 played sessions**.
- Unplayed games are **never factored into the numerator or denominator** (no zero penalties, no default drags).
- If 0 games were played today, the dashboard maintains the patient's calibrated baseline (78) with an explicit notice: *'Baseline 78 (0 games played today) — Unplayed games excluded'*.
- In the 5-domain breakdown, unexercised domains receive score: `null` and are visually badged as *'Excluded / Not played today'*.

The Daily Composite Cognitive Score Formula

Let $S = \{s_1, s_2, ..., s_K\}$ be the set of sessions actually completed today ($K \geq 1$). The composite score synthesizes accuracy, Bayesian latent ability, patient autonomy, and deliberation latency:

1. Mean Accuracy:

2. Mean Latent Ability:

3. Mean Autonomy Score:

4. Mean Reaction Latency:

5. Normalized Theta (0-100):

6. Pacing Deduction:

$Acc_mean = (1 / K) * \sum Acc_k$

$\theta_mean = (1 / K) * \sum \theta_k$

$Aut_mean = (1 / K) * \sum Autonomy_k$

$RT_mean = (1 / K) * \sum Latency_k \text{ (ms)}$

$\theta_norm = clamp(\theta, 100, (\theta_mean + 3.0) * (100 / 6.0))$

$\Delta_latency = (RT_mean > 8000ms ? 5 : RT_mean > 6000ms ? 2 : 0)$

FINAL COMPOSITE SCORE FORMULA:

Daily Composite Score = clamp(50, 99, round(

$0.60 * Acc_mean + 0.25 * \theta_norm + 0.15 * Aut_mean - \Delta_latency$

)))

Dynamic 7-Day Performance Trend Graph Integration

The 7-Day Performance curve displayed in the dashboard is dynamically derived from `DailySessionManager.get7DayPerformanceTrend()`. For days where games were recorded, the curve displays the true composite score of those days' played games. Hovering over any data point reveals an interactive tooltip indicating: *'✓ X games combined (Unplayed excluded)'*, ensuring caregivers and doctors have clear provenance for every score.

3. AI Telemetry Analysis & Dynamic Adaptive Game Flow

The SmritiNER AI architecture operates on two concurrent clinical levels: (1) **On-Device Edge ML Classification** for longitudinal dementia staging, and (2) the **Dynamic Adaptive Game Flow Engine** for intra-session pacing ('changing games on the flow').

A. On-Device Edge ML Classification (Washington University OASIS-2 Trained)

Trained directly on the 371-patient Washington University Alzheimer's Disease Research Center longitudinal dataset, a multinomial Softmax logistic classifier evaluates real-time session telemetry without sending sensitive biometric data to the cloud. With latency <0.2ms, it generates clinical probabilities across 3 diagnostic tiers: **NORMAL** (CDR 0, MoCA 26-30), **MCI** (CDR 0.5, MoCA 18-25), and **HIGH_SUPPORT** (CDR ≥ 1, MoCA < 18).

B. The 5 Dynamic Flow Adaptation Policies ('Changing Games on the Flow')

Policy	Clinical Telemetry Trigger	Dynamic AI Routing Action
1. Hippocampal Relief	Verbal memory strain: Deliberation > 8000ms, Accuracy < 60%, or repeated hints in word-recall or brain-story.	Steers AWAY from verbal recall; routes to Visuomotor Praxis (jigsaw-puzzle) or Wayfinding (where-am-i) to allow temporal lobe recovery.
2. Executive Escalation	High cognitive reserve: Latent ability $\theta \geq +0.85$, Accuracy ≥ 88%, swift reaction (< 4000ms).	Escalates to prefrontal abstraction (odd-one-out) or matrix working memory (pattern-recall) to stimulate neuroplasticity.
3. Motor Stabilization	Parkinsonian tremor / touch jitter index > 0.25 or ≥ 3 debounced tremor taps in fine-motor tasks.	De-escalates from fine drag-and-drop; routes to large-target visual recognition (what-changed, where-am-i) with 400ms filter.
4. Domain Diversification	Multiple exercises completed within the same cognitive domain during today's session.	Prioritizes unexercised DSM-5 neurocognitive domains to ensure holistic balanced stimulation across Memory, Attention, Executive, Spatial, and Language.
5. Palliative Scaffolding	Severe hesitation (> 12,000ms), spatial disorientation, or signs of catastrophic anxiety.	Routes to comforting narrative reminiscence (brain-story Tier 1) with zero time pressure, safeguarding patient dignity and confidence.

4. Unified Clinical PDF Assessment Report Architecture

When a caregiver or physician clicks **'Print / Save PDF'** on the dashboard or inside `ClinicalSessionReportModal.tsx`, the system renders an official, standardized neuropsychological medical document. All gathered session telemetry, domain radar scores, error breakdowns, and ML diagnostic staging are synthesized into a single cohesive clinical dossier.

The 7 Core Sections of the Generated Clinical Report:

1. Patient Demographics & Assessment Metadata

Patient name, chronological age, clinical dementia staging (e.g. MCI), supervising caregiver, attending neurologist, timestamp, and unique cryptographic audit hash.

2. Today's Multi-Game Composite Cognitive Synthesis

The daily composite score (e.g. 84 / 100), mean accuracy (85%), mean latent ability θ (+0.50), mean deliberation latency (3.4s), and the explicit clinical guarantee: *'Unplayed Games Excluded (0 Penalty)'*.

3. Itemized Daily Session Telemetry & Audit Log Table

A complete tabular record of every game played today, itemizing: Protocol Title, Assigned DSM-5 Domain, Accuracy %, Reaction Latency (s), Latent Ability θ , Patient Autonomy %, Error Count, and Clinical Verification Status.

4. DSM-5 Neurocognitive Domain Radar Breakdown

Individual scores across the 5 domains: Delayed Memory Recall, Attention & Working Memory, Executive Flexibility, Visuomotor & Praxis, and Language & Logical Memory. Unplayed domains are explicitly marked 'Excluded / Not played today'.

5. Trained Edge ML Cognitive Staging (OASIS-2 Weights)

Softmax probability distribution across Normal Control, MCI, and Dementia, complete with model confidence percentage and standardized MoCA scale range mapping (e.g. 22 - 25).

6. Neuropsychological Error Pattern Decomposition

Itemized counts for Transposition Errors (DLPFC sequencing check), Intrusion Errors (hippocampal encoding check), and Perseverations (frontotemporal motor inhibition control).

7. Physician & Caregiver Actionable Clinical Guidance

Circadian timing alignment (9:00 AM - 11:30 AM acetylcholine window), tremor accommodation status, hydration prompts, and personalized progression guidelines.

Sample Itemized Multi-Game Audit Table as Rendered in the Clinical PDF:

Protocol / Game	Domain	Accuracy	Reaction Latency	Ability θ	Autonomy	Verification
Smriti Haat (Word Recall)	Language	90%	2.5s (2500ms)	+0.80	100%	✓ Verified
Jigsaw Puzzle (Praxis)	Spatial	80%	4.0s (4000ms)	+0.20	90%	✓ Verified
Sequence Recall (Span)	Attention	85%	3.8s (3800ms)	+0.50	100%	✓ Verified
Odd One Out (WCST)	Executive	—	—	—	—	Excluded (Not Played)
Pattern Recall (Matrix)	Attention	—	—	—	—	Excluded (Not Played)

Clinical Verification & Regulatory Alignment: SmritiNER's telemetry aggregation pipeline adheres strictly to ICMR ethical guidelines and MoHFW digital health standards. Offline edge calculation guarantees zero data leakage while providing neurologists with longitudinal cognitive trajectories.